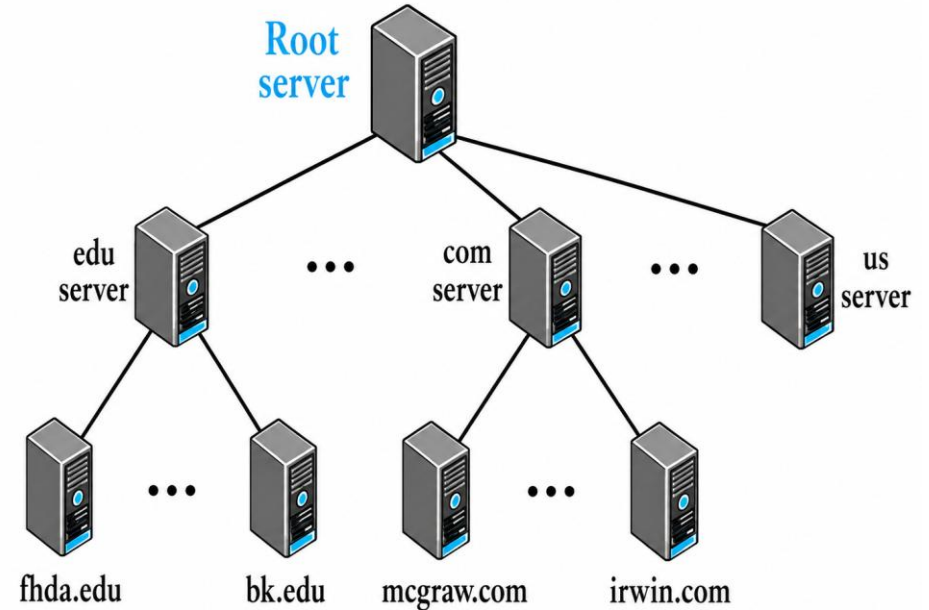
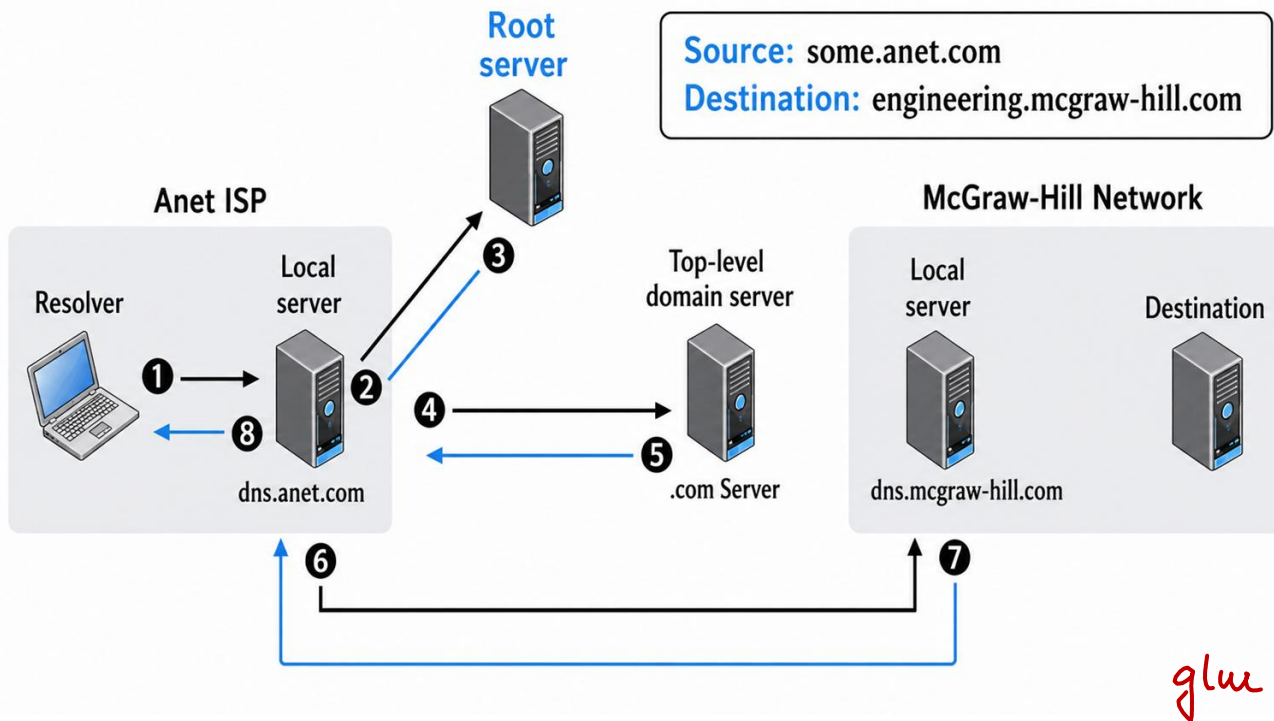


Application Layer Attacks

-Dr. Sumana Maiti

Domain Name Service



To *convert Domain name to IP address*, we need DNS Protocol.

In each of the request, there is an *ID Field*, and in the reply, there is *glue information*.

DNS Attack

Pharming:

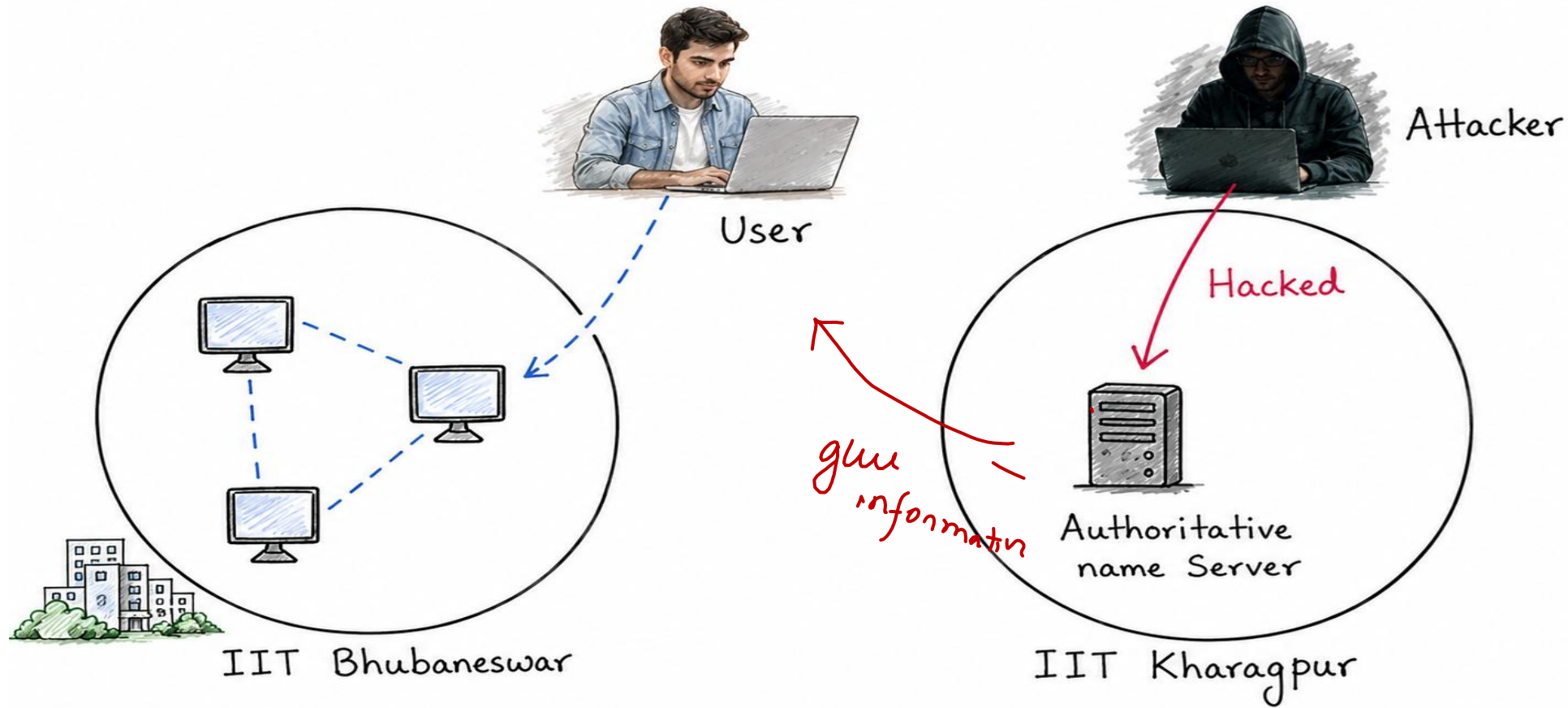
It is an attack where **hostname to IP address mapping** is done maliciously.

hostname → IP address
 ↑
 Attacker's IP address

Phishing:

The **identical website** is run on **attacker's ip address**. Whatever, information the user is sending, it is going to attacker's machine.

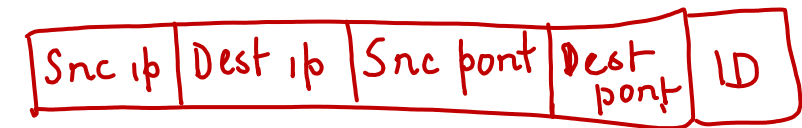
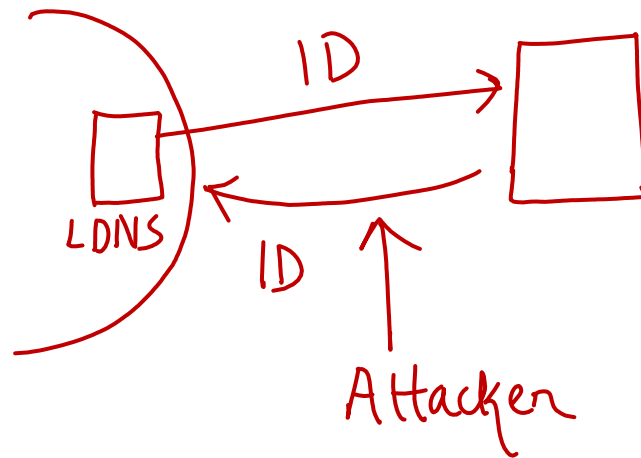
Glue record is used for the **malicious mapping** of IP address



Is there a way that the user is contacting attacker's IP address, when he is accessing IIT Bhubaneswar ?

On-path Attack

- Attacker can see the DNS request.
- Local DNS server is trying to resolve the request of some website mapping.
- The attacker's challenge is to construct the DNS reply in such a way that local DNS server will install the entry.



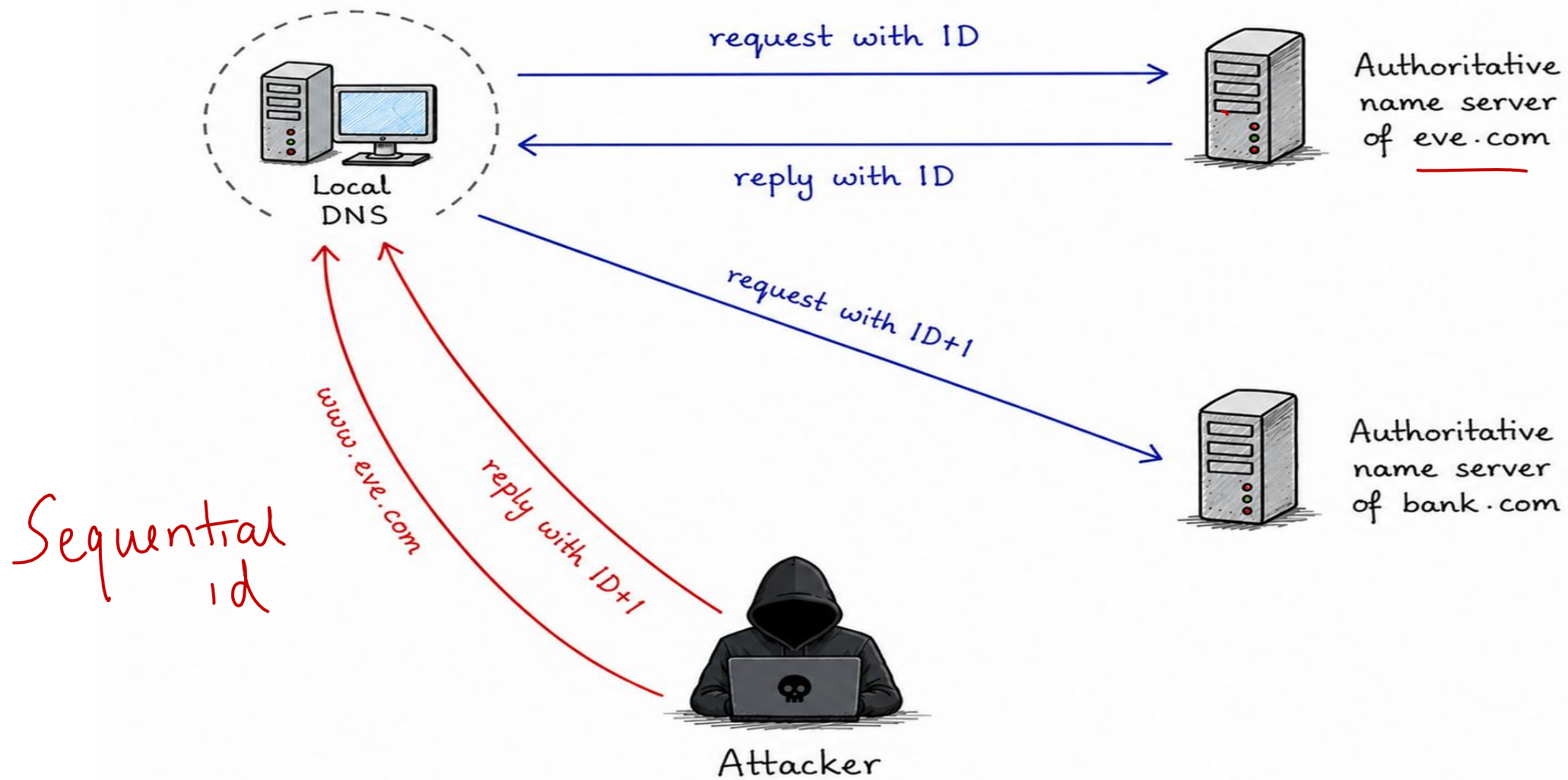
Off-path Attack

- For the attacker, prediction of source port, destination port, source ip, and destination ip is not very hard.
- Attacker's main challenge is to guess the ID as attacker is not able to see the request in this case.
- Without seeing the request, the attacker constructs a reply in such a way that it is accepted by local DNS server.

Example. Attacker asks local DNS server to solve www.eve.com. Local DNS sends the request with some ID and this request will come to attacker's authoritative name server. This authoritative name server is under the control of the attacker. So, attacker knows the id. The request id will be $ID + 1$.

If Sequential ID is used, then this attack can happen. Solution is Random ID

255 50 / ID - 16 bits

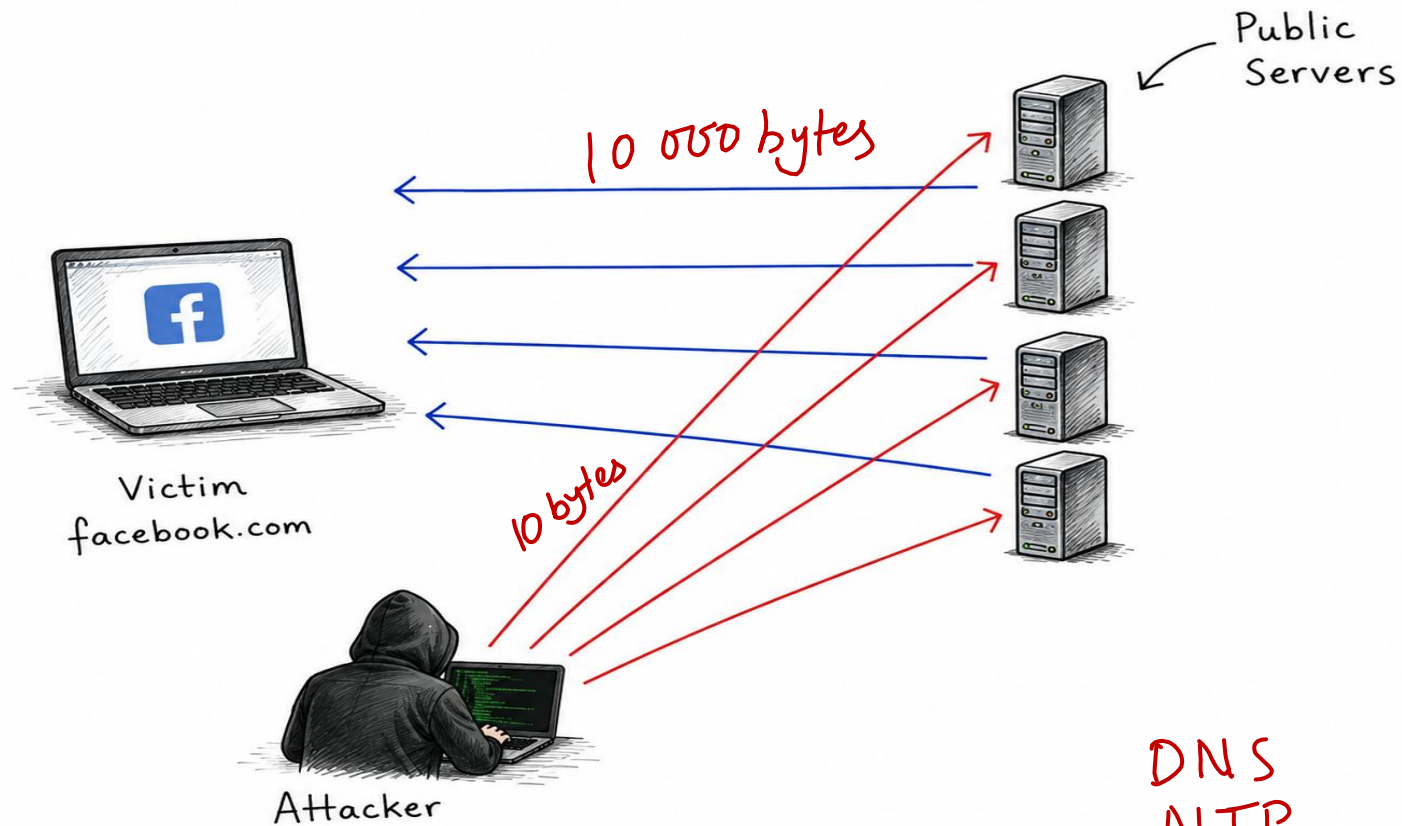


Defense

- Combine the **ID (16 bits)** with **random source port (16 bits)**.
- **DNSSec**: Certificate-based solution.

Digital Signature

Amplification Attack



An **amplification attack** is a type of distributed denial-of-service (DDoS) attack in which an adversary sends small, spoofed requests to third-party servers (Public servers), causing those servers to send much larger responses to a victim, thereby overwhelming the victim's network bandwidth or resources.

Denial of
Service
attack

DNS
NTP